

**Acceptable Means of Compliance and Guidance Material to Annex VI
(Part-ARA) to Regulation (EU) No 1178/2011
Issue 1, Amendment 15**

Annex III to ED Decision 2026/006/R

‘AMC and GM to Part-ARA — Issue 1, Amendment 15’

The amendments are presented as follows:

- deleted text is ~~struck through~~;
- new or amended text is highlighted;
- an ellipsis ‘[...]’ indicates that the rest of the text is unchanged.

As regards Part-ARA, Subpart FSTD, this document shows the AMC and GM that replace the AMC and GM to Part-ARA, Subpart FSTD issued with ED Decision 2012/006/R of 19 April 2012. For these AMC and GM, the presentation provided above is not used.

Note to readers

‘Agency’ and ‘EASA’ are used interchangeably, and this is most visible in consolidated texts. In all cases, both terms refer to the European Union Aviation Safety Agency.

The Annex to Decision 2012/006/R of the Executive Director of the Agency of 19 April 2012 (AMC & GM to Part-ARA — Initial issue) is amended as follows:

ANNEX VI (PART-ARA)

SUBPART GEN – GENERAL REQUIREMENTS

AMC2 ARA.GEN.200(a)(2) Management system

QUALIFICATION AND TRAINING — INSPECTORS

(a) Qualification

[...]

(2) Additional qualification criteria:

[...]

(iv) inspectors approving training programmes should have relevant experience in the same area; ~~and~~

(v) inspectors not involved in activities referred to in (i)-(iv) above should have a relevant background in aviation related to their duties;

(vi) FSTD inspectors should:

(A) in the case of FSTD technical inspectors, hold a qualification as defined in point (a)(1) of AMC2 ARA.FSTD.100(a);(b);(c); and

(B) in the case of FSTD flight inspectors, hold a qualification as defined in point (a)(2) of AMC2 ARA.FSTD.100(a);(b);(c).

(b) Initial training programme

The initial training programme for inspectors should include, as appropriate to their role, current knowledge of, as well as experience and skills in, at least the following:

[...]

(8) criteria for the qualification of FSTDs, including:

(i) assessment and auditing of the organisation operating the FSTD's management systems;

(ii) theoretical knowledge of flight simulation;

(iii) regulatory knowledge related to the FSTD and the organisation operating the FSTD;

(iv) planning and conduct of FSTD evaluations;

(v) verification, validation and testing of the FSTD;

(vi) maintenance and operation of the FSTD; and

(vii) use of the FSTD.

[...]

GM4 ARA.GEN.200(a)(2) Management system

FSTD INSPECTOR TRAINING

(a) The competent authority should cover the training areas as set out in Table 1 below when establishing a training programme for initial and recurrent training for an FSTD inspector to enable the inspector to develop and maintain the required knowledge and skills.

(b) The competent authority may adapt the content of the training areas as necessary, considering:

- (1) the knowledge and experience of the trainee; and
- (2) the tasks and responsibilities of the FSTD inspector.

Table 1 — Training areas for FSTD inspectors

Training area	Scope	Description
Theoretical knowledge of flight simulation	Understanding of flight simulation to be able to assess the functionality, fidelity and compliance of FSTD systems	— FSTD systems and their integration: • Hardware architecture and integration • Flight controls / control loading • Aircraft performance and handling modelling • Flight mechanics and aerodynamics • Avionics software and configuration control • Sound-cueing systems • Motion-cueing systems • Visual systems • Databases and data correlation — Integration of the FSTD systems and their ability to replicate aircraft behaviour — Validation data and validation data roadmaps to FSTD evaluations
Regulatory knowledge	Knowledge of regulatory requirements and procedures to be able to evaluate the compliance and effectiveness of FSTD operations	— International aviation safety regulatory environment (e.g. Chicago Convention on International Civil Aviation and ICAO annexes and documents) — Applicable EU regulations (e.g. Regulation (EU) 2018/1139, Regulation (EU) No 1178/2011) — Acceptable Means of Compliance (AMC) and Guidance Material (GM)) for the application of the applicable EU regulations — Certification specifications for FSTDs — Instructions and best practice guidance (e.g. RAeS, ARINC)
Planning and conduct of evaluations	Planning and conduct of FSTD evaluations, including preparation, execution, reporting and follow-up actions	— Development of evaluation plans and conduct of evaluations in coordination with stakeholders — Review of modifications to FSTDs and records for compliance — Preparation of clear and structured reports — Systematic application of assessment tools and techniques — Compliance and discrepancy identification

Verification, validation and testing	Evaluation of technical and operational performance of FSTD systems through objective and subjective testing and validation data analysis	— Conduct of QTG reviews and scenario-based evaluations — Analysis of validation data and roadmaps — Execution of structured functional tests and identification of performance discrepancies
Maintenance and operation of FSTDs	Assessment of the configuration, maintenance and operation of FSTDs to be able to ensure continued qualification	— Acceptance processes, configuration control and testing — Operational integration of optional systems and impact of technical changes
Use of FSTDs	Understanding of how the FSTD is intended to be used within training programmes, to be able to assess its qualification.	— Use of FSTDs in a training course — Applicable training credits associated with FSTDs — Relevant training, testing and checking related to an FSTD — Operational and flight crew implications of FSTD limitations and capabilities
Assessment and auditing of organisation operating the FSTD's management systems	Assessment of the effectiveness of the management system established by the organisation operating the FSTD in supporting FSTD qualification and compliance	— Audits of compliance monitoring system (CMS), safety management system (SMS) and management processes — Evaluation of system effectiveness and monitoring of follow-up actions

(c) When developing a training programme for initial training of FSTD inspectors, the competent authorities should take into account all the following:

- (1) appropriateness of training areas to different aircraft categories;
- (2) training needs of the inspector to evaluate specific areas related to training (e.g. PBN, UPRT, full stall, HUD, EVS, helicopter-specific scenarios, etc.);
- (3) the individual trainee's development progress when determining the number of audits/evaluations in which the trainee participates as an observer and as a team member under supervision before acting as a full audit/evaluation team member.

(d) The training under supervision in accordance with point (c)(3) should incorporate hands-on evaluation training in an FSTD.

(e) When developing a training programme for recurrent training, the competent authority should include refresher training for FSTD inspectors who do not have recent experience in FSTD evaluations and audits before acting as full audit/evaluation team members.

(f) Any appropriate training methods such as lectures, self-study, web-based training courses, participation in FSTD conferences may be used in the initial and recurrent training of FSTD

inspectors. It is recommended that the trainers of FSTD inspectors have extensive experience in their areas of instruction and adequate andragogical skills.

SUBPART FSTD – SPECIFIC REQUIREMENTS RELATED TO THE QUALIFICATION OF FLIGHT SIMULATION TRAINING DEVICES (FSTDs)

SECTION I – GENERAL

AMC1 ARA.FSTD.100(a) Evaluation procedure

TIMELINE FOR THE INITIAL EVALUATION

- (a) The time allocated to objective tests depends on the speed of the automatic and manual systems set up to run each test and whether or not special equipment is required. The competent authority should not necessarily warn the organisation operating an FSTD of the sample validation tests which should be run on the day of the evaluation, unless special equipment is required.
- (b) The FSTD cannot be used for subjective tests while part of the QTG is being run. Therefore, sufficient time (at least eight consecutive hours) should be set aside for the examination and running of the QTG.
- (c) The subjective tests for the evaluation can be found in the applicable primary reference document (PRD). Note: the suggested subjective test profile is described in GM1 ARA.FSTD.100(a);(b);(c). Depending on the simulated configurations and complexity of the FSTD and simulated aircraft, at least one working day should be planned for the subjective test routine, which effectively prevents the use of the FSTD for any other purpose.
- (d) To ensure adequate coverage of subjective and objective tests and to allow for cost effective rectification and retest before departure of the inspection team, adequate time (approximately three consecutive days) should be dedicated to an initial evaluation of an FSTD.

AMC2 ARA.FSTD.100(a) Evaluation procedure

ASSESSMENT PROCESS LEADING TO THE ISSUE OF AN FSTD QUALIFICATION

- (a) FSTDs require evaluation leading to qualification. The required process should be accomplished in two distinct steps. First, a check should be performed to determine whether or not the FSTD complies with the applicable requirements. When performing this check, the competent authority should ensure that accountability for the issue of an FSTD qualification is clearly defined. In all cases an individual department manager of the competent authority should be appointed under whose personal responsibility the issue of an FSTD qualification is to be considered. The second step should be the grant (or refusal) of an FSTD qualification.
- (b) When checking compliance with the applicable requirements, the competent authority should ensure that the following steps are taken:

- (1) An individual should be nominated by the department manager of the competent authority to oversee, and become the focal point for, all aspects of the FSTD qualification process, and to coordinate all necessary activity. The nominated person should be responsible to the department manager for confirming that all appropriate evaluations/inspections are made.
- (2) The ability of the applicant to secure, in compliance with the applicable requirements and certification specifications, the safe and reliable operation and proper maintenance of the FSTD should be assessed.
- (3) The applicant's proposed management system should be scrutinised with particular regard to the allocated resources. Care should be taken to verify that the system is comprehensive and likely to be effective.
- (4) The competent authority should inform the applicant of its final decision concerning the qualification within 14 days of completion of the evaluation process.
- (5) On completion of the evaluation process, the application, together with a written recommendation and evidence of the result of all evaluations or assessments, should be presented to the nominated person responsible for FSTD qualification. The presentation should be made by the person with overall responsibility, nominated in accordance with (b)(1).
- (6) The department manager of the competent authority should only issue an FSTD qualification certificate if he or she is completely satisfied that all requirements have been met. If he or she is not satisfied, the applicant should be informed in writing of the improvements that are required in order to satisfy the competent authority.
- (7) If an application for an FSTD qualification is refused, the applicant should be informed of the rights of appeal that exist under national regulations.

AMC1 ARA.FSTD.100(a);(b);(c) Evaluation procedure

EVALUATION PROCEDURE — GENERAL

- (a) During FSTD evaluations the competent authority should conduct an appropriate sample of the objective, functions and subjective tests detailed in the applicable qualification basis.
- (b) The competent authority should review an appropriate sample of results of the objective, functions and subjective tests conducted by the organisation operating the FSTD in accordance with points ORA.FSTD.105 (a)(2) and (3).
- (c) For the evaluation of an FSTD, the standard form as mentioned in AMC1 ARA.FSTD.100(d) should be used.

AMC2 ARA.FSTD.100(a);(b);(c) Evaluation procedure

COMPOSITION OF THE EVALUATION TEAM

- (a) The competent authority should appoint a technical team to evaluate an FSTD in accordance with a structured routine to gain or maintain a qualification. The team should consist of at least the following personnel:

- (1) An FSTD technical inspector qualified in aspects of flight simulation, hardware, software, as well as in methods of simulation, with sufficient technical knowledge and experience to understand and analyse objective test results and assess whether the FSTD complies with the applicable certification specifications;

and

- (2) An FSTD flight inspector who holds or has held a privilege to instruct in accordance with Annex 1 to the Chicago Convention and meets either of the following:
 - (i) for the aircraft being simulated, holds or has held the relevant class or type rating, as applicable; or
 - (ii) is assisted by a person from the organisation operating the FSTD with qualifications as specified in point (a) of AMC1 ORA.FSTD.100(a).

In any case, a pilot from the organisation operating the FSTD as referred to in point (a) of AMC1 ORA.FSTD.100(a) should support the technical team appointed by the competent authority to evaluate the FSTD.

- (b) An FSTD technical inspector or an FSTD flight inspector should be:
 - (1) staff of the competent authority performing the evaluation; or
 - (2) accredited inspector(s) from another competent authority; or
 - (3) person(s) designated by the competent authority.
- (c) If an FSTD inspector has not participated in an FSTD evaluation in the last three years, such inspector should undergo recurrent training with a focus on areas as relevant for the FSTD evaluation, based on the needs of the individual inspector.
- (d) On a case-by-case basis for recurrent evaluations, the competent authority may reduce the evaluation team to one inspector, provided that all of the following conditions are met:
 - (1) such an evaluation team composition is not being used prior to the second recurrent evaluation;
 - (2) an evaluation performed solely by one inspector is followed by an evaluation with an evaluation team, composed by an FSTD technical inspector and an FSTD flight inspector;
 - (3) the inspector satisfies the criteria for both positions as referred to in points (a)(1) and (2);
 - (4) no major modification has been implemented since the preceding evaluation performed by the competent authority;
 - (5) no relocation of the FSTD has taken place since the last evaluation;
 - (6) a system is established enabling the competent authority to monitor and analyse the status of the FSTD on a continuous basis;
 - (7) the FSTD hardware and software has been working reliably for the previous years, which should be reflected in the number and kind of discrepancies recorded in technical log entries and the results of the compliance monitoring system audits.
- (e) In the case of a BITD, the recurrent evaluation may be conducted by one suitably qualified FSTD flight inspector only.

AMC3 ARA.FSTD.100(a);(b);(c) Evaluation procedure

FUNCTIONS AND SUBJECTIVE TESTS — SUGGESTED TEST ROUTINE

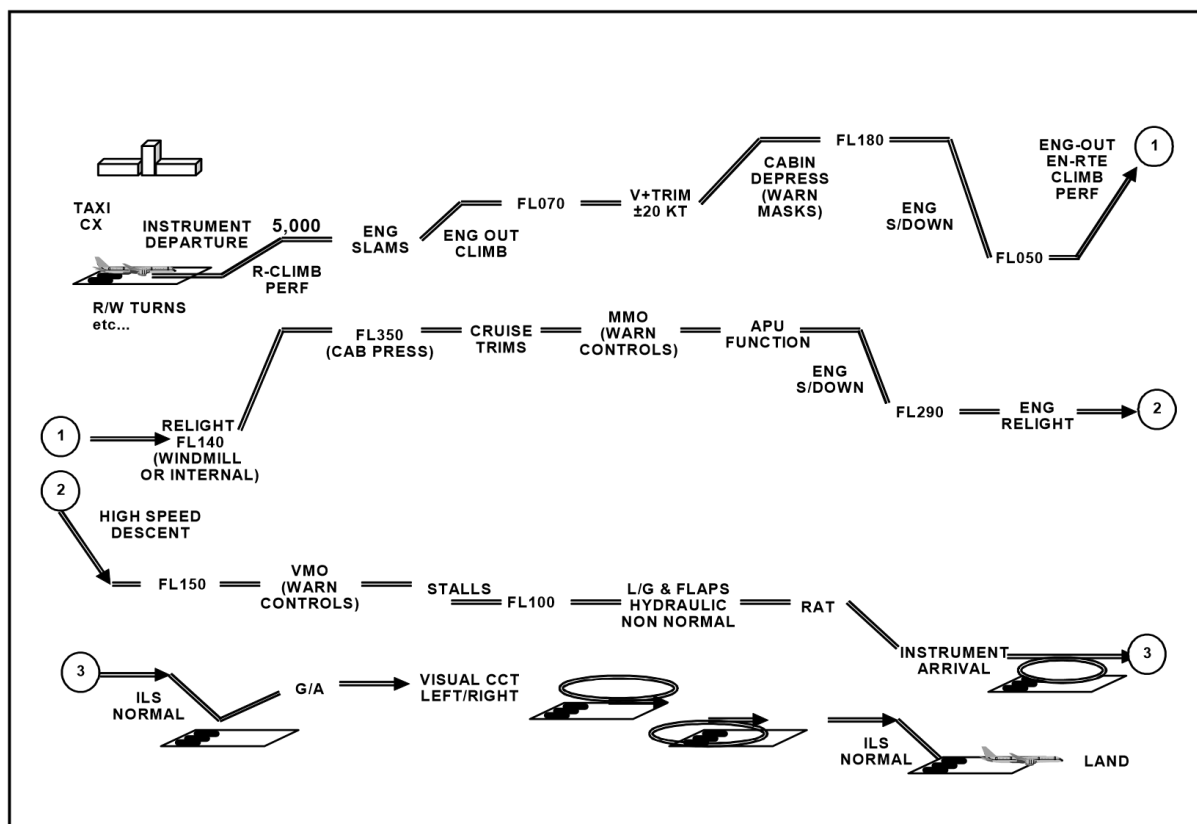
- (a) During initial and recurrent evaluations of an FSTD, and when deemed relevant in the event of a major modification to an FSTD, the competent authority should conduct a series of functions and subjective tests that together with the objective tests complete the comparison of the FSTD with the type, variant or group of aircraft, as applicable.
- (b) The FSTD should provide adequate flexibility to permit the accomplishment of the desired and required tasks while maintaining an adequate perception by the flight crew that they are operating in a real aircraft environment. Additionally, the instructor operating station (IOS) should not present an unnecessary distraction from observing the activities of the flight crew whilst providing adequate facilities for the tasks.
- (c) The competent authority should perform sampling of the functions and subjective tests that are applicable to the device. Part of the subjective test routine for an FSTD should involve an uninterrupted fly-out and a flight where the FSTD controls such as flight freeze and repositioning are used. The evaluator(s) should brief the persons participating in the evaluation on the planned flight contents and working methods.

GM1 ARA.FSTD.100(a);(b);(c) Evaluation procedure

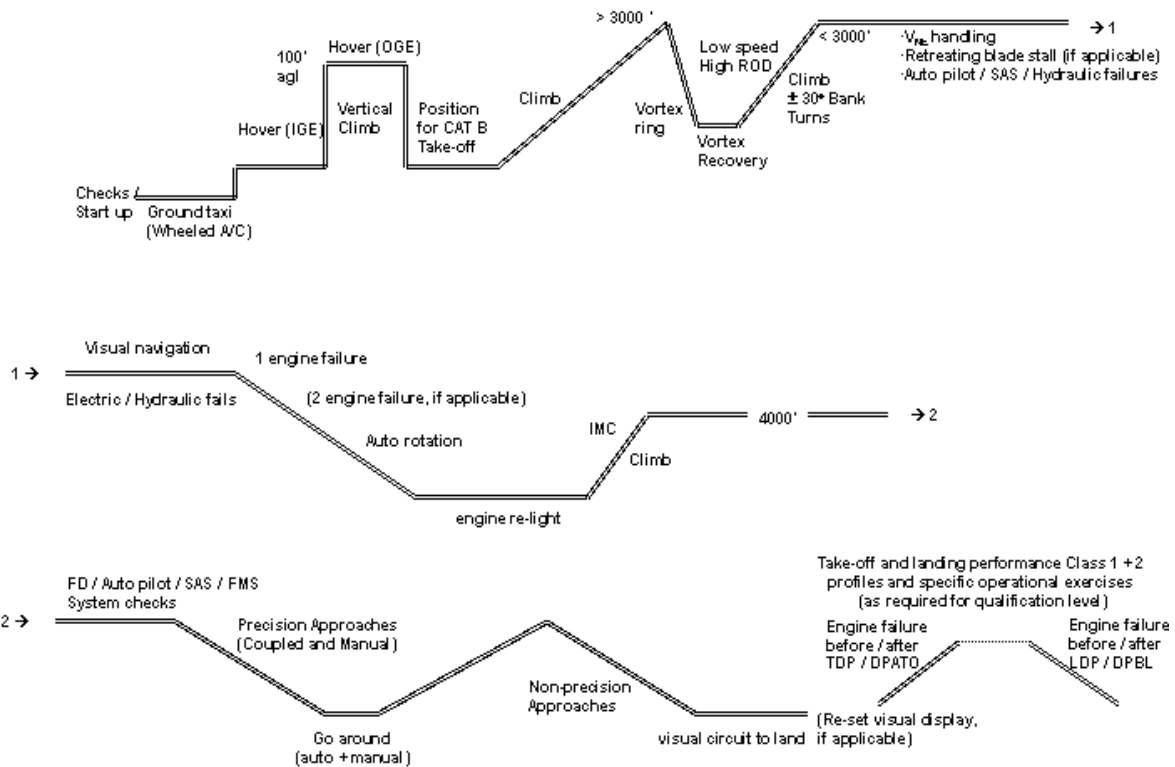
TYPICAL TEST PROFILES

The uninterrupted fly-out may be most efficiently performed by preparing a flight test profile. Examples of such profiles are provided below and should be adapted for each FSTD.

- (a) Typical test profile for an FSTD aeroplane:



- (b) Typical test profile for an FSTD helicopter



AMC1 ARA.FSTD.100(d) Evaluation procedure

FSTD EVALUATION REPORT

Logo of the Competent Authority

[competent authority] FSTD EVALUATION REPORT

Date of the report
Report issue number
Organisation operating the FSTD
[Competent Authority] FSTD identification code:
Aircraft type, variant or group of aircraft (only for legacy FSTDs):
Engine fit(s) simulated:

Contents

1. Flight simulation training device (FSTD)
2. Evaluation details
3. Supplementary information
4. FSTD qualification
5. Classification of items
6. Results
7. Evaluation team

The conclusions presented are those of the evaluation team. The competent authority reserves the right to change these after internal review.

1. Flight simulation training device (FSTD)				
(a) Organisation operating the FSTD:				
(b) FSTD location:				
(c) (Competent Authority) FSTD identification code:				
(d) FSTD manufacturer and FSTD identification serial number:				
(e) First entry into service (month/year):				
(f) Visual system (manufacturer and type):				
(g) Motion system (manufacturer and type):				
(h) Engine instrumentation: Flight instrumentation:				
2. Evaluation details				
(a) Date of the evaluation:			(b) Date of the previous evaluation:	
(c) Type of evaluation: initial recurrent special				
(d) Technical criteria primary reference document:				
(e) Validation data roadmap (VDR) ID-No:				
3. Supplementary information				
(a) Evaluation team				
	Name	Position	Organisation	
Technical inspector				
Flight inspector				
Support pilot				
Support staff				
(b) Visual databases used during the evaluation:				
(c) Software load reference used during evaluation:				
(d) ESL revision No and date:				
(e) Other:				
4. FSTD qualification				
(a) FSTD qualification level (recommended) (only for legacy FSTDs):				
(b) FSTD qualified for MCC training (recommended):				
(c) FSTD qualified for UPRT (approach to stall or full/post stall) (recommended):				
(d) Recommended limitations:				
FCS recommended				
	FSTD FEATURE		FIDELITY LEVEL	Simulated aircraft
1.	Flight deck layout and structure (FDK)			

2.	Flight control forces and hardware (CLH)		
3.	Flight control systems operation (CLO)		
4.	Aircraft systems (SYS)		
5.	Performance and handling on ground (GND)		
6.	Performance and handling in-ground effect (IGE)		
7.	Performance and handling out-of-ground effect (OGE)		
8.	Sound cueing (SND)		
9.	Vibration cueing (VIB)		
10.	Motion cueing (MTN)		
11.	Visual cueing (VIS)		
12.	Navigation (NAV)		
13.	Atmosphere and weather (ATM)		
14.	Operating sites and terrain (OST)		

5. Classification of items

During the FSTD evaluation, various identified items may necessitate operator resolution. It is crucial to differentiate between two distinct categories when managing these items: severity A and severity B.

Severity A item

Items categorised as severity A by the evaluation team are non-compliances related to one of the following:

- a simulation feature, system, subsystem, component or part that fails to comply with the qualification basis, which may affect the FCS or the assigned FCS, the ESL or the FSTD qualification, and training, testing and checking and/or potentially lead to a negative transfer of training;
- non-compliance with the qualification basis, such as objective test results being out of tolerance.

If a limitation is to be added to the qualification certificate, it should be entered as 'Severity A item' with the text 'Limitation:' preceding the item description.

Severity B item

Items categorised as severity B by the evaluation team are non-compliances related to simulation features, systems, subsystems, components, or parts that do not meet the conditions for classification as a severity A item.

Observation

The evaluation team records an observation in cases where compliance with the required standard is not clearly proven and therefore a subsequent assessment by the FSTD organisation is necessary. Following the outcome of each assessment, the competent authority may promote an 'Observation' to 'Item' and inform the organisation operating the FSTD.

Period of rectification

The status of each 'Item' and 'Observation' should be notified to the competent authority within the period specified in point ARA.FSTD.100(e)(1). Failure to address or clarify the 'Item' may

result in an enforcement measure by the competent authority in accordance with point ARA.FSTD.135.

6. Results

6.1 Severity A item

1	
2	

6.2 Severity B item

1	
2	

6.3 Observation

1	
2	

6.4 Comments

1	
2	

7. Evaluation team, designated by the competent authority

Signed For the competent authority

GM1 ARA.FSTD.100(d) Evaluation procedure

FSTD EVALUATION REPORT — GUIDANCE ON CLASSIFICATION OF NON-COMPLIANCES (ITEMS)

- (a) When providing an evaluation report to the organisation operating the FSTD, the competent authority should classify the non-compliances as either severity A items or severity B items.
- (b) When classifying non-compliances as severity A items, the following examples may be considered. This list is not exhaustive and severity A items need to be classified on a case-by-case basis:
 - (1) subjective tests that deviate from the aircraft behaviour and have adverse impact on training, testing, checking or potentially lead to negative training if the device is used in such condition;
 - (2) one or several QTG is/are out of tolerance;
 - (3) the check for the relevant systems has failed;
 - (4) recurring, systemic or critical defects.

- (c) When classifying non-compliances as severity B items, the following examples may be considered. This list is not exhaustive and severity B items need to be classified on a case-by-case basis:
- (1) simulation features, systems, subsystems, components or parts that are temporarily inoperative, unusable or operating below their nominal level;
 - (2) limited use of simulation features, systems, subsystems, components or parts is possible.
- For severity B items, the overall serviceability of the FSTD is demonstrated by the organisation.

AMC1 ARA.FSTD.115 Interim FSTD qualification

NEW AIRCRAFT PROGRAMME

- (a) Aircraft manufacturers' final data for performance, handling qualities, systems or avionics is seldom available until well after a new or derivative aircraft has entered service. Because it is often necessary to begin flight crew training and certification several months prior to the entry of the first aircraft into service, it may be necessary to use aircraft manufacturer-provided preliminary data for interim qualification of FSTDs.
- (b) In recognition of the sequence of events that should occur and the time required for final data to become available, the competent authority may accept the use of certain preliminary aircraft and systems data, and early release avionics in order to permit the necessary programme schedule for training, certification and service introduction.
- (c) The use of preliminary data is not indefinite. The aircraft manufacturer's final data should be available within six months after the aircraft's first 'service entry' or as agreed by the competent authority, the organisation and the aircraft manufacturer, but usually not later than one year. When an organisation applies for an interim qualification using preliminary data, the organisation and the competent authority should agree upon the update programme. This update programme should normally specify that the final data update will be installed in the FSTD within a period of six months but not later than two years following the final data release, unless special conditions exist, and a different schedule has been agreed between the organisation operating the FSTD and the competent authority. If the final data is not implemented in the subject FSTD within 36 months from its initial qualification, the competent authority should act in accordance with point ARA.FSTD.135(b).
- (d) For the qualification of FSTDs of new aircraft types, using the services of a suitably qualified test pilot for the purpose of assessment of handling qualities and performance evaluation may be beneficial.

AMC1 ARA.FSTD.120 Continuation of an FSTD qualification

GENERAL

- (a) *Objective Testing.* During recurrent evaluations, the competent authority should seek evidence of the successful running and analysis of the QTG between evaluations. The competent authority should select a number of tests to be run during the evaluation, including those that may be a cause for concern. Adequate notification should be given when special equipment is required for the test.

- (b) Essentially the time taken to run the objective tests depends on the need for special equipment, if any, and the test system, and the FSTD cannot be used for subjective tests or other functions whilst testing is in progress.
- (c) For a modern FSTD incorporating an automatic test system, four hours would normally be required. FSTDs that rely on manual testing may require a longer period of time.
- (d) *Functions and subjective testing.* A functions and subjective test profile should be flown to sample a selection of the subjective tests, as appropriate, based on the past performance of the device and the applicable qualification basis. Examples of a test profile which may be performed as part of the uninterrupted fly-out are provided in GM1 ARA.FSTD.100(a);(b);(c).
- (e) Normally, the time taken for recurrent subjective testing is about four hours, and the FSTD should not perform other functions during this time. If additional configurations (e.g. second engine or avionics fit) are to be tested, more time should be planned as necessary.
- (f) To ensure adequate coverage of functions and subjective tests and objective tests during a recurrent evaluation, a total of eight hours should be allocated. However, it should be noted that any FSTD discrepancy that arises during the evaluation could necessitate the extension of the duration of the testing.

AMC2 ARA.FSTD.120 Continuation of an FSTD qualification

EXTENDED EVALUATION PERIOD — GENERAL

- (a) The competent authority should determine the length of the extended recurrent evaluation period of an FSTD, taking into consideration the information listed in point (a) of AMC2 ORA.FSTD.225(b). The extended evaluation period should be 24 or 36 months.
- (b) The extension of the recurrent evaluation period should be driven by a risk-based approach as part of the continuous oversight. The performance of the organisation, as well as the performance of the device, should be monitored.
- (c) Once the competent authority has granted an extended recurrent evaluation period, it may plan all subsequent recurrent evaluations of the relevant FSTD with the extended time interval of 24 or 36 months, as applicable, until it decides to revert again to a shorter recurrent evaluation period, should the conditions of point ARA.FSTD.120(c) be no longer fulfilled.
- (d) The audit of the management system elements, described in point ARA.FSTD.120(c)(3), may be performed remotely, at the discretion of the competent authority, considering the performance of the organisation.

AMC1 ARA.FSTD.130 Modifications

GENERAL

- (a) When considering applications for a modification of an FSTD, the competent authority should ensure that accountability for the modification is clearly defined.
- (b) A member of the competent authority should be appointed under whose personal authority an FSTD qualification may be modified.

- (c) On receipt of an application for a modification of an FSTD, the competent authority should conduct such evaluations and inspections as are necessary to ensure that the full implications of the request have been addressed by the organisation operating the FSTD.
- (d) During the processing of a modification request, the continued adequacy of the management system should be reviewed.
- (e) When the request has been considered and examined, the competent authority should decide on the depth of inspection of the FSTD that is required.
- (f) The appointed member of the competent authority, if satisfied that the organisation operating the FSTD remains competent and the qualification of the FSTD can be maintained, should issue a revised FSTD qualification certificate, as appropriate.
- (g) The competent authority should inform the organisation operating the FSTD of its decision within 30 days of receipt of all documentation where no evaluation is required, or within 14 days of any subsequent evaluation.

AMC2 ARA.FSTD.130 Modifications

EVALUATION CONSIDERATIONS FOR FSTD MODIFICATIONS

- (a) In order to determine the need for a special evaluation, the competent authority should consider:
 - (1) the nature of the modification, its novelty, its complexity and the risk associated with the modification;
 - (2) the performance of the organisation operating the FSTD.
- (b) The competent authority should consider the following factors when deciding on the scope and schedule of a special evaluation:
 - (1) the FSTD modifications to be assessed using the applicable qualification basis;
 - (2) the impact of the changes to the ESL;
 - (3) the impact of the changes to the MQTG, such as:
 - (i) updated validation data in the validation data roadmap;
 - (ii) updated validation tests and tolerances;
 - (iii) updated statements of justification.
- (c) If the organisation operating the FSTD applies for a major modification while maintaining the assigned FCS of the FSTD as specified in point (f)(2)(i) of AMC1 ORA.FSTD.110, the competent authority should conduct an evaluation in accordance with point ARA.FSTD.100(c) to assess the FSTD using its original qualification basis. If such major modification results in lowering the capabilities of the FSTD as established with the original qualification basis, the competent authority should act in accordance with point ARA.FSTD.100(e).
- (d) If the organisation operating the FSTD applies for a major modification to obtain an FCS as specified in point (f)(2)(ii) of AMC1 ORA.FSTD.110, the competent authority should conduct an evaluation in accordance with point ARA.FSTD.100(c) to assess the FSTD using the applicable qualification basis at the time of application for the major modification. Before re-issuing an FSTD qualification certificate with the new FCS, the competent authority should inform the applicant about the result of the evaluation.

GM1 ARA.FSTD.130 Modifications

SPECIAL EVALUATION CONSIDERATIONS FOR FSTD WITH FCS

AIRCRAFT SIMULATION FEATURES

- (a) The aircraft simulation group of FSTD simulation features comprises, amongst others, 'Performance and handling on ground', 'Performance and handling in-ground effect', 'Performance and handling out-of-ground effect', 'Flight control forces and hardware' and 'Flight control systems operation'. Changes to fidelity levels of these features usually directly affect handling and performance qualities and will result in updates to the QTG that may justify an in-depth special evaluation covering objective as well as subjective testing.
- (b) Aircraft simulation also includes the features 'Flight deck layout and structure' as well as 'Aircraft systems'. Changes to fidelity levels of these features certainly require functions and subjective testing but may still require updates to the QTG documentation for general requirements and statements of justification, even if there are no objective validation tests affected.

CUEING SIMULATION FEATURES

- (c) The cueing simulation FSTD group of features comprises 'Visual cueing', 'Motion cueing', 'Vibration cueing' and 'Sound cueing'. Changes to fidelity levels of all these cueing features will likely impact the QTG general and objective testing requirements and will require both objective and subjective testing at the discretion of the competent authority. Some visual cueing feature updates may comprise mirror replacement and/or projector replacements for which the competent authority may not require a special evaluation unless the FCS is affected.

ENVIRONMENT SIMULATION FEATURES

- (d) The environment simulation FSTD group of features comprises 'Navigation', 'Atmosphere and weather' as well as 'Operating sites and terrain'. Changes to these feature fidelity levels will probably only require functional and subjective testing.

AMC1 ARA.FSTD.135 FSTD qualification certificate – limitation, suspension, revocation

FSTD QUALIFICATION CERTIFICATE ENFORCEMENT — MEASURES

- (a) The competent authority's inspection and monitoring process should confirm the competent authority's continued confidence in the effectiveness of the management system of the organisation operating the FSTD, and its ability to maintain an adequate standard.
- (b) If the organisation operating the FSTD fails to satisfy the competent authority's concerns, a written notification should be given to the organisation together with a firm date by which specified action to satisfy the competent authority should be taken. It should be made clear that failure to comply may result in enforced limitation, suspension or revocation of the FSTD's qualification.
- (c) The competent authority should preserve quality of training, testing and checking which is of paramount importance and therefore the competent authority may immediately limit, suspend or revoke any FSTD qualification which it has issued.

AMC2 ARA.FSTD.135 FSTD qualification certificate – limitation, suspension, revocation

SUSPENSION AND LIMITATION

- (a) When a decision has been taken to suspend or limit an FSTD qualification certificate, the organisation operating the FSTD should be informed immediately by the fastest available means. The phrase ‘quickest available means’ will in most situations mean the use of a facsimile or email message.
- (b) In the event of a full suspension of an FSTD qualification certificate, the organisation operating the FSTD should be instructed that the FSTD concerned cannot be used for any credited training, testing or checking.
- (c) Following the submission of the facsimile or email message as referred to in point (a), the competent authority should send to the organisation operating the FSTD a formal letter giving notice of suspension or limitation and setting out the conditions on which suspension or limitation may be lifted.
- (d) If the competent authority is notified by the organisation operating the FSTD about the cessation of the FSTD operations for a period in excess of six months, the competent authority should consider opening the process described in point (b) of AMC1 ARA.FSTD.135.
- (e) The FSTD qualification certificate should not remain suspended indefinitely. Further steps may be taken by the organisation operating the FSTD to reinstate the FSTD qualification or, in default, by the competent authority to revoke the FSTD qualification certificate. Should an organisation operating the FSTD wish to dispute the suspension of its FSTD’s qualification certificate, it should be informed of its rights of appeal that exist under national regulations. If an appeal is lodged, the FSTD qualification may remain suspended until the appeal process is complete.
- (f) Suspension of an FSTD qualification certificate may be lifted on appeal or if the organisation operating the FSTD restores the FSTD to its previously acceptable standard.
- (g) In neither case should operations be permitted to restart until it has been demonstrated that the cause of the suspension or limitation has been rectified. The competent authority may require a special evaluation depending on the severity of the problem.
- (h) The competent authority should issue a formal notice of the lifting of suspension before the organisation operating the FSTD is permitted to resume use of the FSTD.

AMC3 ARA.FSTD.135 FSTD qualification certificate– limitation, suspension, revocation

REVOCATION

- (a) The competent authority should give the organisation operating the FSTD a notice that it intends to revoke the FSTD qualification followed by a formal letter of revocation.
- (b) Should the organisation operating the FSTD wish to dispute this revocation, it should be informed of its rights of appeal as exist under applicable regulations. Once revoked, there can be no further activities under the terms of the FSTD qualification.